Ex: 9

TEXT GENERATION USING LSTM NETWORKS

Problem Statement :

Build a text generation model using Long Short-Term Memory (LSTM) networks. Train the model on a text corpus to generated coherent sequences of text and evaluate the output for fluency and coherence.

Objectives :

1. Understand sequential modeling for natural language generation.
2. Train a character-level LSTM model to learn language patterns.
3. Generate text using a seed prompt and evaluate the results.
4. Analyze the fluency and creativity of LSTM generated outputs.

Implementation Steps :-

Step 1 : Load and preprocess the Dataset

```
import tensorflow as tf
import numpy as np
text = tf.keras.utils.get_file('shakespeare.txt')
text = open(text, 'rb').read().lower()
chars = sorted(set(text))
c2i = {c:i for i, c in enumerate(chars)}
i2c = {i:c for i, c in enumerate(chars)}
```

Step2: Create Input and output sequences

```
seq_len = 40
x = []
y = []
for i in range (len (text) - seq_len):
    input_seq = text [i: i + seq_len]
    target_char = text [i + seq_len]
    x.append ([[c2i[c] for c in input_seq])
    y.append (c2i[target_char])
x = np.array (x)
y = np.array (y)
```

Step3: Build the LSTM model

```
model = tf.keras.Sequential ([
    tf.keras.layers.Embedding (len (chars), 64, input_
                               length = seq_len),
    tf.keras.layers.LSTM (128),
    tf.keras.layers.Dense (len (chars), activation =
                           'Softmax')
])
model.compile (loss = 'sparse_categorical_crossentropy',
               optimizer = 'adam')
model.fit (x, y, batch_size = 128, epochs = 1)
```

Step4: Define the Text Generation Function

```
def generate (seed, length = 300):
    seq = [c2i[c] for c in seed.lower ()]
    for _ in range (length):
        inp = np.array (seq[-seq_len:]).reshape(1,-1)
        pred = model.predict (inp, verbose = 0)[0]
        next_idx = np.random.choice (len (pred), p=pred)
        seq.append (next_idx)
    return seed + "".join ([i2c[i] for i in seq[len (seed):]])
```

Generated Text:

shall i compare thee to a summer's day?

solko:
though with you, thomme, thy cousserarse on.

vunlulibe:
why wenges with but proor?

sonromala:
and and sthen morivinario are, by your up
but you enoul thy speiring; won hanger:
there wink gress'd has my vurifory with:
o, lo, will supk onn name, to soul hongait;
'tis rome beance shibben thea

Step5: Generate and display Text

```
print ("\n Generated Text:\n")
print (generated ("Shall I compare thee to a summer's
                   day? \n"))
```

RESULT:

The LSTM model learned LSTM model learned character-level text patterns and generated shakespearean-style text, demonstrating sequence learning and text generation capabilities.